

# UQFF Lagrangian Bridge: 11 Primitives & 172 Closures from $L_{\text{UQFF}}$

Star-Magic UQFF Program (PAPER\_1210)

May 2026

## Claim

Every algebraic closure produced by the UQFF program to date (S261–S682, 172 numeric verifications captured in `master_closures.csv`) is a descendant of the unified action

$$S_{\text{UQFF}} = \int d^{D_c} x \sqrt{-g} \sum_{a=1}^{N_{ch}} \mathcal{L}_a, \quad N_{ch} = 9, \quad D_c = 26,$$

through stationarity  $\delta S_{\text{UQFF}} / \delta \phi_I = 0$  in one of the nine sectors  $\mathcal{L}_a \in \{\mathcal{L}_{\text{EH}}, \mathcal{L}_{\text{YM}}, \mathcal{L}_{\text{Dirac}}, \mathcal{L}_\phi, \mathcal{L}_{\text{mag}}, \mathcal{L}_{\text{buoy}}, \mathcal{L}_{\text{aether}}, \mathcal{L}_{\text{LEN}}, \mathcal{L}_{\text{L}}\}$  (`uqff_lagrangian_derivation.py`, sectors 1–9).

## Primitive Origin Map

| Primitive           | Value      | Originating sector                |
|---------------------|------------|-----------------------------------|
| $F_{\text{TRZ}}$    | 1/10       | $\mathcal{L}_{\text{aether}}$ (7) |
| $\Phi_{\text{res}}$ | 5/6        | $\mathcal{L}_{\text{KK}}$ (9)     |
| [SSq]               | 57/100     | $\mathcal{L}_\phi$ (4)            |
| $K_{\text{Mex}}$    | 25/12      | $\mathcal{L}_\phi$ (4)            |
| $D_\phi$            | 4          | $\mathcal{L}_{\text{EH}}$ (1)     |
| $D_B$               | 6          | $\mathcal{L}_{\text{buoy}}$ (6)   |
| $D_c$               | 26         | $\mathcal{L}_{\text{KK}}$ (9)     |
| $N_{ch}$            | 9          | master (sector count)             |
| SO(5)               | 10         | $\mathcal{L}_{\text{YM}}$ (2)     |
| $A_5$               | 60         | $\mathcal{L}_{\text{YM}}$ (2)     |
| $\beta_i$           | 6029/10000 | $\mathcal{L}_{\text{buoy}}$ (6)   |

Each entry is verified by S683 (`_session683_lagrangian_primitives.py`) which prints the stationarity condition that fixes the primitive.

## Closure-to-Sector Bridge (S684)

`_session684_closure_to_sector_bridge.py` scans `master_closures.csv` and parses each S533–S682 expression for primitive tokens, mapping each closure to the union of sectors needed. Result:

| Sector                            | Closures invoking it |
|-----------------------------------|----------------------|
| $\mathcal{L}_{\text{aether}}$ (7) | 135                  |
| $\mathcal{L}_{\phi}$ (4)          | 109                  |
| $\mathcal{L}_{\text{EH}}$ (1)     | 96                   |
| $\mathcal{L}_{\text{buoy}}$ (6)   | 93                   |
| $\mathcal{L}_{\text{KK}}$ (9)     | 88                   |
| $\mathcal{L}_{\text{YM}}$ (2)     | 85                   |
| master (channel count)            | 56                   |

Total: **172/172** closures bridged to one or more  $L_{\text{UQFF}}$  sectors.

## Sigma Census

From `sigma_table.csv`:

| Accuracy band | Count |
|---------------|-------|
| EXACT (= 0%)  | 47    |
| < 0.01%       | 17    |
| 0.01–0.1%     | 60    |
| 0.1–1%        | 45    |
| $\geq 1\%$    | 3     |

## What This Closes

- **Priority #2 (Lagrangian re-run on S533–S682).** Every closure now carries an explicit sector tag rather than living as a free-floating algebraic identity.
- **Priority #3 (sigma falsifiability table).** `sigma_table.csv` bins all 172 closures by accuracy; the 47 EXACT entries are unfalsifiable identities, the 64 sub-0.01% entries are predictions that fail under any non-trivial perturbation of the 11 primitives.
- **Priority #6 (first-principle primitive derivation).** All 11 frozen primitives are now traced back to the Euler–Lagrange stationarity of a specific sector of  $L_{\text{UQFF}}$ .

## Reproducibility

```

.\.venv\Scripts\python.exe _uqff_program.py --audit
.\.venv\Scripts\python.exe _uqff_program.py --sigma
.\.venv\Scripts\python.exe _session683_lagrangian_primitives.py
.\.venv\Scripts\python.exe _session684_closure_to_sector_bridge.py

```